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Characteristic	Number of patients (%)
Tumor location	
• Anterior	52 (12)
• Posterior	366 (88)
T stage	
• T1	88 (21)
• T2	159 (37)
• T3	142 (34)
• T4	20 (5)
• Unknown	9 (3)
COMS	
• Small	38 (9)
• Medium	257 (62)
• Large	118 (28)
• Unknown	5 (1)
Visual acuity at diagnosis (in 298 patients)	
• Full loss	1 (1)
• p-ph	59 (20)
• mps- 0.1	62 (21)
• 1- 0.4	61 (20)
• 4- 0.6	62 (21)
• 0.7 and higher	53 (17)

rates of OS and DMFS were similar with ≥ 45 Gy and < 45 Gy but 2- and 5-year EFS rates were significantly higher with ≥ 45 Gy (82% and 66% vs. 76% and 50% with < 45 Gy, $p = 0.006$). The rates of OS, DMFS, and EFS were significantly lower in patients with a tumor diameter ≥ 8 mm. In patients with a tumor depth ≥ 8 mm, the rates of DMFS and EFS were significantly lower. In patients with a tumor volume < 600 mm³, the rates of OS, DMFS, and EFS were significantly higher. Among patients that received ≥ 45 Gy, the ones with a tumor volume < 600 mm³ had significantly higher rates of DMFS and EFS but OS was similar. In multivariate analysis, no prognostic factors were found for DMFS. However, for OS and EFS > 8 mm basal tumor diameter (relative risk [RR] = 1.6, 95% confidence interval [CI] = 1.92-3.12, $p = 0.09$; and RR = 1.8, 95% CI = 1.15-2.81, $p = 0.01$), for EFS > 8 mm depth (RR = 1.6, 95% CI = 1.03-2.4, $p = 0.03$), and ≤ 45 Gy dose (RR = 1.4, 95% CI = 0.9-2.08, $p = 0.06$) were negative prognostic factors. SRS/FSRT-related late toxicity was significantly higher in patients with a basal tumor diameter > 8 mm and depth > 8 mm ($p = 0.001$ and $p = 0.04$, respectively).

Conclusion: Total SRS/FSRT dose and tumor diameter and depth were found prognostic factors for EFS. Toxicity rate was higher in patients with a higher tumor depth and basal diameter.

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Effect Of Timing Of Adjuvant Radiosurgery On Local Control For Brain Metastases



D.B. Vanderbilt,¹ D.A. Julie,¹ A. An,¹ P. Christos,² S. Pannullo,¹ J.P.S. Knisely,¹ and T.H. Schwartz¹; ¹Weill Cornell Medical College/New York-Presbyterian Hospital, New York, NY, ²Division of Biostatistics and Epidemiology, Department of Healthcare Policy and Research, Weill Cornell Medical College, New York, NY

Purpose/Objective(s): Surgical resection of brain metastasis from solid tumors improves overall survival and functioning, and adjuvant radiotherapy with stereotactic radiosurgery (SRS) may be employed for high rates of local control (LC). The optimal timing of adjuvant SRS following surgical resection of brain metastases is uncertain. This retrospective study examined the effects of the timing of postoperative SRS on disease recurrence.

Materials/Methods: Patients were identified ($n = 133$) having undergone resection of solid tumor brain metastasis with adjuvant SRS. The time from surgery to SRS was compared by Wilcoxon two-sample test. The local recurrence-free survival (LRFS) of patients having SRS within 4 weeks of surgery was compared to those who had SRS at an interval greater than 4 weeks using a Log-rank test. A logistic regression model was constructed to evaluate predictors of local recurrence (LR).

Results: Median tumor size was 2.9 cm, and 84% of patients received gross total resection (GTR). The most common histology was non-small cell lung cancer. The median time from surgery to SRS was significantly longer for patients who had LR (58.7 days) as compared to patients with no LR (37.9 days, $p < 0.0001$). For those who had GTR, a longer median time from surgery to SRS was also observed for patients with LR (54.7 days) compared with patients without LR (38.4 days, $p = 0.0007$). For patients having SRS within 4 weeks of surgery, there was significantly improved LRFS compared to patients with SRS delayed beyond 4 weeks ($p < 0.01$, median LRFS not reached for either group). There was a trend towards higher local recurrence for tumors larger than 3 cm ($p = 0.0633$), and for patients with GTR, tumor size > 3 cm was a significant predictor of LR ($p = 0.0495$). Age, performance status, tumor location, and pathology were not significant predictors of LR in this cohort.

Conclusion: A longer interval between surgery and adjuvant SRS for brain metastasis of solid tumors may be associated with local recurrence, even among patients with GTR.

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Low Dose ¹²⁵I Brachytherapy And Transpupillary Thermal Therapy (TTT) For Small To Medium Size Choroidal Melanoma

S. Fattahi,¹ K.S. Corbin,² C.L. Deufel,² K.V. Astafurov,³ T.W. Olsen,³ S.L. Stafford,² M. Neben Wittich,² J. Viehman,⁴ L.A. Dalvin,³ and I.A. Petersen²; ¹Alix School of Medicine, Mayo Clinic, Rochester, MN, ²Department of Radiation Oncology, Mayo Clinic, Rochester, MN, ³Department of Ophthalmology, Mayo Clinic, Rochester, MN, ⁴Division of Biomedical Statistics and Informatics, Mayo Clinic, Rochester, MN

Purpose/Objective(s): To evaluate our institutional outcomes after treatment of small to medium size choroidal melanoma with low dose COMS brachytherapy.

Materials/Methods: Single center, retrospective review of small to medium size choroidal melanoma < 5 mm in apical height managed from 2005-2019 with ¹²⁵I brachytherapy at prescribed dose < 80 Gy, using a homogeneous dose calculation methodology. Ocular outcomes included visual acuity, presence of ocular radiation side effects, and local tumor recurrence. The Kaplan-Meier method was used to estimate overall survival, and metastasis rates were reported using cumulative incidence with

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Patient Characteristics				
Median age	61 years (IQR: 51-68 years)			
Sex	Female 58%			
Comorbidities	Atherosclerotic disease 44%	Diabetes 18%	Hyperlipidemia 45%	Hypertension 49%
Tumor Characteristics				
Median logMAR visual acuity	0.18 (IQR: 0.0-0.4)			
Snellen acuity	20/15-20/40 68%	20/50-20/160 29%	20/200 or worse 3%	
Tumor dimensions	Largest basal diameter (IQR) 8.0 mm (IQR: 6.7-10.0 mm) Thickness (IQR) 2.0 mm (IQR: 2.0-2.0 mm)			
Secondary retinal detachment	18%			
Subfoveal/subretinal fluid	88%			